

Channel-Aware Task-Oriented Semantic Granularity Selection for Bandwidth-Constrained V2V Cooperative Perception

Peiyi Yue

Abstract—Connected and automated vehicles can improve perception reliability by exchanging information through vehicle-to-vehicle (V2V) links. However, cooperative perception must operate under limited bandwidth and time-varying vehicular channels. Existing communication-efficient perception methods mainly reduce redundancy within a fixed semantic representation, such as object-level detections or feature-level messages. This paper shows that fixed semantic granularity is suboptimal: feature-level messages are potentially more informative but can become unreliable or wasteful under poor channel quality, while object-level messages are compact and stable but less expressive. We propose Channel-Aware Task-Oriented Semantic Granularity Selection (CA-TOSG), a receiver-driven framework in which the ego vehicle selects, for each frame, one granularity from the deployed action set $\mathcal{S} = \{E, L, F\}$ —ego-only, object-level, or feature-level (the denser 256-QAM feature mode is a physical-layer comparator, not a deployed action)—from its own perception cues and estimated channel state, and signals the choice to the collaborator via a 2-bit request piggy-backed on periodic V2X awareness messages, such as CAM/BSM, over 802.11bd or NR sidelink. CA-TOSG chooses one message type from object-level communication and compressed feature-level communication modes before any high-payload transmission. We derive the selector as the Lagrangian-relaxed solution of a constrained perception-communication problem and instantiate it with a 400-tree Random Forest. Experiments on the OPV2V dataset (validate, scene-disjoint test, and the Culver-City domain shift) under AWGN and Rayleigh channels report the realised true end-to-end AP@0.5 of the frozen selectors descriptively across three prescribed average communication budgets $B_{\max} \in \{0.10, 0.20, 0.30\}$. How much AP any granularity policy can contest on a split is bounded by that split’s headroom, the gap between the perfect-channel feature-level ceiling and fixed object-level communication: 0.0550 on validate, 0.0240 on test and 0.0970 on Culver-City. Read the same way on every split, the frozen selector converts at most about a fifth of that headroom (up to 21.3%, and as little as 0.0% on Culver-City at the tightest budget), and we claim no AP advantage on any split. What the selector does buy is channel use: averaged over all channel states it spends 3.7–21.4% of the per-frame channel use of fixed feature-level transmission under rate-1/2 LDPC + 16-QAM coding, and under fading it falls back to the robust object-level message. The channel-type and estimated SNR features dominate the selector’s feature importance, jointly accounting for 61.7% against 38.3% spread over 21 ego-side cues — importance, not sufficiency: with the channel features alone the selector stops requesting features at the tighter budgets. The result is therefore not an accuracy result: a one-line SNR threshold and a three-scalar hand rule reach comparable F1, but they do so by sending more feature messages, and what the task uses

buy is knowing which of the frames the channel can carry are actually worth sending. The single pre-registered confirmatory comparison is against the nominal SNR-threshold rule on the scene-disjoint test split at $B_{\max} = 0.20$: the selector is non-inferior within the pre-registered 0.005 margin while reducing communication payload by 34.8%, a comparator that is itself over budget at $0.2168 > 0.20$ Msym. Against the budget-matched π_{feasible} the selector is ahead on F1 by +0.00067 at 26.6% lower payload; that comparison is secondary and no non-inferiority decision is attached to it. Neither half of the input is sufficient on its own: given only the channel state the selector stops requesting features altogether at the two tighter budgets, and under deep fading the digital feature branch turns infeasible over the evaluated SNR range. Whether a graceful codec shifts that balance is exploratory evidence only, reported in an appendix under the prior protocol and not re-evaluated under the frozen one. The deployed selector incurs 52.1 ms per frame on a single CPU core (the slowest of the three frozen selectors), fitting the 100 ms budget of a 10 Hz LiDAR cycle. These results demonstrate that channel-aware semantic granularity adaptation is an effective strategy for bandwidth-constrained V2V cooperative perception.

Index Terms—Vehicle-to-vehicle communication, cooperative perception, connected vehicles, semantic communication, 3D object detection, channel-aware communication.

I. Introduction

Connected and automated vehicles can exchange perception information over vehicle-to-vehicle (V2V) links to obtain complementary viewpoints, which matters most exactly where a single vehicle’s LiDAR or camera is incomplete: intersections, dense traffic, occlusions and long range. Cooperative perception can therefore improve detection robustness beyond the sensing range and viewpoint of one vehicle [1], [2].

It is, however, constrained by communication. Raw point clouds or dense feature maps preserve the most information but can exceed what vehicular links carry under bandwidth and latency limits, and V2V channels are time-varying through mobility, blockage, multipath and fading [3], [4]: a strategy that works on a clean channel can be unreliable at low signal-to-noise ratio (SNR). Cooperative perception must therefore account for the reliability and cost of the link, not only for perception.

Existing methods sit at different points of that trade-off. Late fusion transmits object-level predictions [1]: compact and strongly protectable, but discarding semantic context and weak when local detections are uncertain. Intermediate fusion transmits learned feature maps [2], [5]:

Peiyi Yue is with the Department of Electrical and Electronic Engineering, University of Bristol, Bristol, UK (e-mail: ppyue9836@gmail.com).

richer, at a much higher communication cost, which recent work reduces by selecting important regions, masking or compressing [5], and which semantic communication and joint source-channel coding optimise for task performance under wireless channels [6], [7].

All of them, however, work at a fixed semantic granularity: object-level methods always send object-level messages, feature-level methods always send feature-level ones, even when only a subset of tokens is selected. They answer which feature elements to transmit, not the prior question of whether feature-level communication is needed for this frame at all—and in some frames local perception is already confident enough for the compact message, while in occluded or ambiguous frames the richer one helps. The best granularity is not fixed; it depends on the task state.

The communication channel further changes this decision. Feature-level messages are potentially more informative, but they are also more sensitive to channel degradation because larger and denser messages require more reliable transmission. Under poor SNR or fading, a feature-level message may be corrupted or effectively unusable, making its additional payload wasteful or even harmful for task performance. In contrast, object-level messages are much smaller and can be treated as robust low-rate information in many system designs. As a result, richer semantic communication does not always lead to better cooperative perception. The optimal message granularity depends jointly on perception task cues and channel quality.

Motivated by this observation, this paper proposes Channel-Aware Task-Oriented Semantic Granularity Selection (CA-TOSG) for bandwidth-constrained V2V cooperative perception. Instead of transmitting a fixed representation for every frame, CA-TOSG selects one message type from a set of semantic communication candidates. In the considered implementation, the selector chooses between ego-only operation, a compact object-level message and a compressed feature-level message, which differ substantially in per-frame channel use (a denser 256-QAM feature mode is defined but excluded as dominated; Section III-B). The decision is made using ego-side task cues, such as prediction confidence, visibility, temporal stability, and detection uncertainty, together with estimated channel quality such as SNR. The selected message is then transmitted through the V2V link and fused with the ego vehicle's local feature for 3D object detection. An overview of the resulting framework is shown in Fig. 1.

This formulation is complementary to existing feature compression and semantic communication methods. A feature codec, sparse feature selector [5], or joint source-channel coding module [6], [7] can be used inside the feature-level branch. CA-TOSG operates one level above these modules by deciding when feature-level communication should be activated. This is especially relevant for vehicular networks, where unnecessary high-payload communication should be avoided when the channel cannot support reliable feature transmission or when the current

frame does not require additional semantic detail.

The main contributions of this paper are summarized as follows:

- We formulate channel-aware semantic granularity selection for V2V cooperative perception, where the ego receiver adaptively selects the semantic level it needs and requests it, according to its own task state and estimated channel quality; the collaborator transmits the requested level.
- We instantiate the framework as a lightweight, interpretable, receiver-driven policy that uses only ego-side perception cues and an estimated SNR/channel-type, adds no learnable parameters to the perception backbone, and whose inference fits within the 100 ms frame interval of a 10 Hz LiDAR cycle on a single CPU core (we measure the selector, not the end-to-end chain). The policy is deliberately model-agnostic: a 400-tree Random Forest [8] and a one-line SNR threshold are two policy realisations with different payload-F1 operating points, so the contribution is the selection framework, not a particular classifier.
- We provide a communication-aware evaluation protocol under AWGN and Rayleigh channels. The mainline comparison is between fixed ego-only, object-level and feature-level transmission with LDPC + 16/256-QAM coding [3], an SNR-threshold rule at both its nominal and its budget-matched tuning, two- and three-scalar hand rules over the same cues, and a channel-aware oracle. A learned semantic-communication baseline [6] is reproduced as an exploratory codec comparison under the prior protocol and is reported in the supplementary material, not in the mainline tables.
- Experiments on OPV2V show that CA-TOSG avoids unreliable high-payload feature transmission under poor channel conditions and approaches oracle performance when the channel supports richer communication. We characterise where it sits on the payload-F1 plane rather than claiming it is better on average, and report, per budget, the comparison against an SNR threshold tuned on validate for the same target budget, and show that, under a frozen selector, the communication saving transfers to the scene-disjoint test split and to the Culver-City domain shift (payload reductions of 32.2–49.2% and 42.1–68.9% against the nominal threshold, which is itself over budget at $B_{\max} = 0.20$ and 0.30), while the F1 non-inferiority does not extend to Culver-City at $B_{\max} = 0.20$, where the selector falls 0.00883 below the threshold rule (95% CI $[-0.00902, -0.00865]$), outside the pre-registered 0.005 margin.
- Under the evaluated LDPC + QAM transport, channel state carries most of the granularity decision — most, not all, and only on this transport: neither half of the input is sufficient on its own, and given only the channel state the selector stops requesting features at the two tighter budgets. Under deep

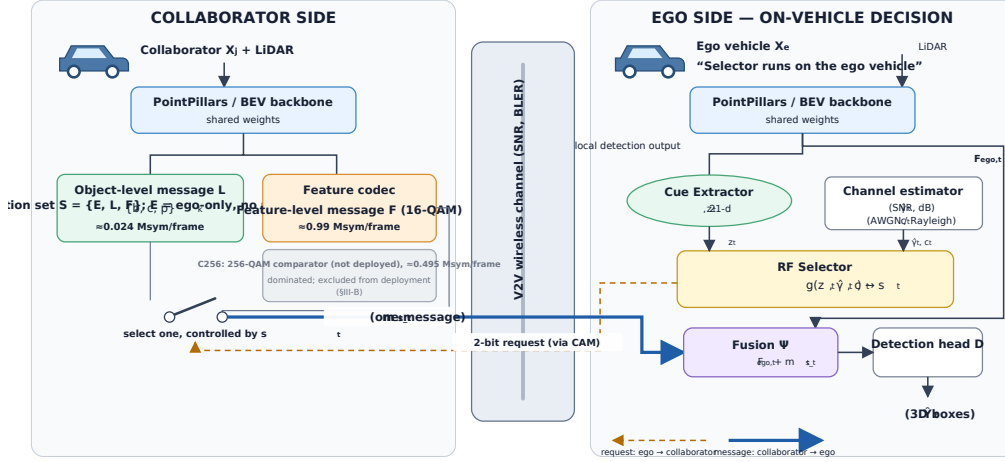


Fig. 1. Overview of the proposed CA-TOSG framework. The ego vehicle extracts online task cues z_t and estimated channel quality (γ_t, ch_t) , then selects one communication mode $s_t \in S = \{E, L, F\}$, where E is ego-only operation with no request at all (the dominated C_{256} mode is a physical-layer comparator and is not in the deployed action set, Section III-B). Only the selected message is transmitted through the V2V link and fused with the local ego feature $F_{\text{ego},t}$ for 3D object detection.

fading the codec choice instead sets feasibility: the digital LDPC feature branch is infeasible over the evaluated SNR range, its frame-error threshold rising as the diversity order falls. Whether a graceful codec changes that currency is exploratory evidence only (the supplementary material); it was not re-evaluated under the frozen protocol and no claim in this paper rests on it.

II. Related Work

A. V2V Cooperative Perception

V2V cooperative perception exploits information from surrounding connected vehicles to alleviate occlusion, long-range sparsity and limited field of view [1], [9]. Methods are categorised by the stage at which information is shared.

Early fusion shares raw point clouds before feature extraction [1]: the most complete information, but a very large payload requiring accurate spatio-temporal alignment, which is hard to deploy under practical V2V bandwidth and latency constraints.

Late fusion shares object-level predictions—boxes, classes, confidences [1]. The payload is small and suits low-rate links, but intermediate semantic and spatial information is discarded: an object missed by the sender cannot be recovered by the receiver, which costs accuracy in occluded or uncertain scenes.

Intermediate fusion shares learned BEV features between backbone and detection head [2], [5], [10]–[15], balancing the two extremes: richer spatial and semantic information than object-level sharing, but messages far larger, and expensive under bandwidth-constrained links.

Benchmarks such as OPV2V and OpenCOOD [1] provide common datasets and protocols for comparing these strategies. We build on that setting but ask a

different question: rather than assuming a fixed fusion representation, how should the semantic granularity of the transmitted message be selected per frame from task and channel conditions?

B. Communication-Efficient Cooperative Perception

To reduce the communication burden of intermediate fusion, many methods aim to transmit only task-relevant features. A common strategy is to use confidence maps, attention weights, or importance masks to select informative spatial regions from the feature map [5], [6]. For example, spatially selective methods transmit only a subset of BEV tokens that are expected to contribute to downstream detection. This reduces redundant communication while retaining much of the benefit of feature-level cooperation.

Where2comm [5] is a representative method in this direction. It estimates where communication is needed and selectively transmits spatial feature regions for cooperative perception. Such methods are effective because not all BEV locations contribute equally to object detection. Attention-based and confidence-based approaches follow a similar principle: they suppress unimportant feature regions and allocate communication resources to more informative areas.

Feature compression is another important direction. Instead of directly transmitting dense feature maps, the sender can encode them into a lower-dimensional representation before communication [6]. The receiver then decodes or fuses the compressed feature representation for detection. These approaches can reduce payload and may be combined with channel coding or learned semantic communication modules.

Most of these methods, however, optimise within a fixed semantic level: they assume feature-level transmission and choose which tokens or compressed representations to

send. The closest exception is ML-Cooper [16], whose soft actor–critic policy adapts the proportion of raw-, feature- and object-level data to the network state—a heavyweight learned policy that does not characterise when granularity adaptation is required. Existing approaches either mix collaboration levels by channel condition [16], adapt content within the feature level [5], [6], or choose hybrid fusion by spatial overlap [17]. CA-TOSG instead makes a lightweight, interpretable, receiver-driven per-frame choice across object- and feature-level representations, from ego-side task cues and explicit channel reliability, under a practical digital PHY with delivery-failure fallback.

C. Semantic and Channel-Aware Communication for Vehicular Perception

Semantic communication optimises the transmitted representation for downstream task performance rather than for reconstruction or bit error rate [6], [7], [18]–[20]; in cooperative perception it should preserve what detection needs, not the full feature map. A recent multi-mode IoT variant jointly selects the transmission mode and allocates resources across sensors [21]—a different axis from per-frame object-versus-feature granularity for V2V perception.

Joint source-channel coding and learned semantic codecs provide one way to improve robustness under noisy wireless channels [6], [7]. By training the encoder and decoder with channel effects in the loop, these methods can produce representations that degrade more gracefully than conventional separated source and channel coding under certain channel conditions. Digital communication baselines, such as modulation and coding schemes based on QAM and LDPC, often show threshold or waterfall behavior [3]: performance can be poor below a certain SNR and improve rapidly once the channel becomes reliable.

For vehicular perception, this channel dependence is especially important. V2V links experience time-varying SNR, multipath propagation, blockage, and fading [3]. A feature-level message that is useful under a clean channel may become unreliable under low SNR or Rayleigh fading. Therefore, the semantic value of a transmitted message depends not only on its content but also on whether the channel can support its reliable delivery.

Some make that transmission channel-adaptive at a fixed granularity: SmartCooper [22] adjusts the feature compression ratio from the estimated channel state, AccBEV [23] conditions a feature-repair module on the SNR. Both adapt how the feature message is encoded or repaired while still committing to feature-level transmission, so they are complementary: such a codec can sit inside CA-TOSG’s feature branch, because CA-TOSG acts before the mode is chosen.

A concurrent line of work pushes semantic communication for collaborative perception into the digital domain. CoDS [24] pairs a task-oriented semantic compression codec with a semantic analog-to-digital converter, so that learned features traverse a standard digital V2X

bitstream – LDPC-coded and QAM-modulated, the same cliff-prone class of digital transport we characterise – rather than an analog channel. On the sender side a feature-selection network identifies task-important spatial locations for transmission; on the receiver side an uncertainty-aware network discards features corrupted by decoding failures, mitigating the LDPC cliff after decoding. CoDS thus selects by task importance at the sender and filters by decode reliability at the receiver. CA-TOSG makes the prevailing channel state a first-class conditioning signal alongside the task: from local task cues and the estimated channel state it selects the message granularity per frame – ego-only operation with no request at all, the compact object-level message, or the feature-level message – and signals the choice through the 2-bit request before any high-payload transmission; when the channel cannot carry a feature message, the feature-level actions are gated out of the feasible set and the selector defaults to the compact object-level message. The two are complementary rather than competing – a digital semantic codec such as that of CoDS could serve as the feature-level branch that CA-TOSG activates, placing channel-conditioned granularity selection one level above codec and spatial-selection design. Both address the cliff effect, at different points: CoDS discards corrupted features after they are received, whereas CA-TOSG avoids spending the collaborator’s transmission budget on an undeliverable feature message. This channel-aware semantic granularity selection is particularly suitable for vehicular systems, where communication resources are limited and channel quality can change rapidly.

III. System Model and Problem Formulation

A. Cooperative Perception Setting

We consider a single ego–collaborator V2V link. At each frame t , the ego vehicle observes the scene using its onboard sensors and extracts a local bird’s-eye-view (BEV) feature representation, denoted by $F_{\text{ego},t}$. The collaborating vehicle also observes the scene from its own viewpoint and extracts a local feature representation $F_{j,t}$ using a shared PointPillars backbone [25]. The collaborator can transmit a perception message to the ego vehicle through the V2V link. The ego vehicle then fuses the received message with its local feature and produces the final 3D object detection output.

We adopt a receiver-driven signalling architecture: the ego vehicle evaluates its own perception state and channel state, decides which semantic granularity is needed, and signals the choice to the collaborator using a 2-bit request piggy-backed on the existing Cooperative Awareness Message (CAM), carried over IEEE 802.11bd or NR sidelink [3]. The request thus costs 2 bits per frame at 10 Hz, i.e., 20 bps, which is negligible relative to the 24 kbit per frame (0.24 Mbit/s at 10 Hz) L payload and is already provisioned in the standard CAM signalling budget. The 2-bit field carries four codepoints; the deployed instance uses three— E (ego-only, no request), L and F —leaving

the fourth codepoint and the non-deployed C_{256} comparator unused. The collaborator complies by transmitting the requested message m_t^{st} . This receiver-driven design makes all perception cues used by the selector locally observable at the ego (Section IV-B) and avoids any causality issues that would arise if the collaborator had to estimate ego-side perception state without feedback.

Let Y_t denote the ground-truth objects at frame t , and let $\hat{Y}_t$ denote the predicted detection output at the ego vehicle after fusion. The objective of cooperative perception is to improve the task utility of $\hat{Y}_t$, such as average precision (AP) or F1 score, while keeping the V2V communication payload low. Unlike conventional cooperative perception settings that assume a fixed type of transmitted representation, we consider an ego-side selector that can choose the semantic granularity of the requested message according to both perception-side task cues and the estimated channel condition.

B. Message Candidates

At each frame, the ego receiver selects one communication mode from the deployed action set and requests it

$$s_t \in \mathcal{S} = \{E, L, F\}, \quad (1)$$

where E denotes ego-only operation, L object-level communication, and F the compressed feature-level message, transmitted under 16-QAM modulation with rate-1/2 LDPC coding; we write C_{16} for the same message where the modulation itself is the subject.

E is a first-class action, not merely a failure mode. Choosing E means the ego requests nothing and detects from its own LiDAR alone, so it costs $B_E = 0$ channel uses and cannot fail in transit. It is the right choice on two distinct kinds of frame: those where the ego's own detection is already sufficient, so any message would buy nothing, and those where the channel is so poor that a feature request would almost surely be lost and an object-level message would still cost bandwidth for little gain. E also serves as the delivery-failure fallback: when a requested F message does not arrive, the receiver falls back to the ego-only detection, which is why the effective utility of F in Eq. (6) interpolates between the feature-level and the ego-only result according to the block-error probability. The two roles share one quantity but are decided at different times— E as an action is chosen before transmission by the selector, whereas E as a fallback is what remains after a delivery failure.

The same feature-level message is additionally evaluated with 256-QAM (C_{256} , same rate-1/2 LDPC) as a physical-layer comparator, but it is not included in the deployed semantic action set $\mathcal{S}$: modulation order is a transport parameter, not a semantic granularity (Section III-B). The transmitted message is therefore

$$m_t = m_t^{st}, \quad (2)$$

where only the message corresponding to the selected mode s_t is transmitted. This one-of-many design is important: the sender does not transmit both object-level

TABLE I
Notation for the semantic granularity levels. F and C_{16} denote the same message; the C_q form is used only where the modulation order is the subject.

Symbol	Meaning	Channel use	Role
E	ego-only, no transmission	$B_E = 0$	deployed action; also the delivery-failure fallback
L	object-level message	$B_L = 0.024$	deployed action
$F \equiv C_{16}$	feature-level, 16-QAM	$B_F = 0.99$	deployed action
C_{256}	feature-level, 256-QAM	0.495	physical-layer comparator, not deployed

and feature-level messages simultaneously, but chooses the semantic granularity that is expected to provide the best task-communication trade-off for the current frame.

The object-level message is defined as a compact set of local detections:

$$m_t^L = \{(b_k, c_k, p_k)\}_{k=1}^{N_t}, \quad (3)$$

where N_t is the number of detected objects at the sender, b_k is the 3D bounding box, c_k is the class label, and p_k is the confidence score of object k .

The compressed feature-level message is defined as

$$m_t^{C_q} = \phi_q(F_{j,t}), \quad q \in \{16, 256\}, \quad (4)$$

where $\phi_q(\cdot)$ denotes a feature encoding and communication process for mode C_q . The two feature-level modes carry the same perception payload of approximately 1.98 Mbit but require different numbers of channel uses because of the different spectral efficiency of 16-QAM versus 256-QAM.

Of the two feature-level modes, the 256-QAM variant (C_{256}) carries the identical semantic content as C_{16} —the same compressed feature payload of Eq. (7)—and differs only in modulation order, which halves the per-frame channel use (0.495 vs. 0.99 Msym for C_{256} and C_{16} ¹). It is included as a physical-layer comparator: holding the semantic content fixed and changing only the constellation is what isolates the codec's channel response in Fig. 2. It is not an element of the deployed action set $\mathcal{S} = \{E, L, F\}$ (Eq. (1)), and it is excluded by design rather than by measurement: the selector chooses what to send—a semantic granularity the receiver can request with the 2-bit field—whereas modulation order is a transport parameter of the link, not a granularity. Fig. 2 shows why the distinction is physically consequential: the 256-QAM frame-error cliff sits far to the right of the 16-QAM one (AWGN onset 16.5 vs. 8.0 dB²), and under Rayleigh and OFDM both flatline at the ego floor across 0–20 dB, so on this transport a denser constellation buys channel use at the cost of reach.

¹Feature payload 1.98 Mbit (Eq. (7)) at LDPC rate-1/2 gives 3.96 Mbit of coded bits; $\div 8$ bit/sym (256-QAM) = 0.495 Msym, $\div 4$ bit/sym (16-QAM) = 0.99 Msym.

²Es/N0 onset at which the frame-level BLER first falls below 0.999 – the same 0.999 constant that defines the feasibility mask. Fig. 2 plots both cliffs directly.

C. Channel Model

Let γ_t be the estimated channel quality at frame t in dB, available from pilot-based estimation, link-layer feedback or RSRP reports as already produced by IEEE 802.11bd and 5G NR sidelink stacks [3]; it is an explicit input to the selector. Three channel models are used: AWGN, which is noise-limited and gives a controlled SNR-dependent setting; Rayleigh, which captures multipath fading; and a frequency-selective OFDM link over a tapped-delay-line channel with an exponential power-delay profile at $\tau_{\text{rms}} = 46.2$ ns (3GPP TR 37.885 Urban NLOSv [4]), whose cross-subcarrier LDPC coding carries frequency diversity; a log-log slope fit of its codeword BLER over the waterfall region gives an empirical diversity order of 2.01, versus ≈ 1 for flat Rayleigh and the sharp cliff of AWGN. All per-frame block-error rates are tabulated from a Sionna link-level simulation. The feature-level message under mode C_q traverses the channel as a coded LDPC block, with block-error rate (BLER) function $\text{BLER}_q(\gamma)$ that is monotonically non-increasing in γ and exhibits the threshold behavior characteristic of LDPC decoding. This is an all-or-nothing full-frame delivery model: a frame's feature message is usable only if all N_{cw} of its constituent codewords decode, with no codeword-level retransmission, erasure protection, or HARQ. We bound what that assumption costs rather than only naming it: fragmentation alone leaves the frame BLER algebraically unchanged, and one retransmission per fragment improves the marginal AWGN cliff point while leaving the Rayleigh branch closed across the evaluated range (Table VI). The Rayleigh infeasibility is therefore a property of the link budget rather than of the no-HARQ assumption, while the AWGN thresholds would move under partial recovery. Carrying the modified block-error rates and the inflated payload $B_F(1 + q_{\text{first}})$ through the full deployment replay leaves every conclusion in place: over all three splits, all three budgets and $k \in \{1, 2, 4\}$, one retransmission moves the selector by at most $+0.00064$ F1 and $+0.00803$ Msym, and the SNR-threshold rule not at all at $B_{\text{max}} = 0.20$, because its feature requests already sit above the cliff (results/sensitivity/r25_fragmentation_replay.csv). We treat the object-level message L as a low-rate robust message in the evaluated SNR range, motivated by its small payload (approximately 0.024 Mbit) that allows aggressive channel coding or repeated transmission. Fig. 2 plots the resulting $\text{BLER}_q(\gamma)$ for the LDPC + 16/256-QAM configurations under both channels, making explicit the threshold (cliff) behaviour that the selector must track.

D. Effective Task Utility under Channel Corruption

Let F_t^s denote the realized frame F1 of the ego vehicle's detection output when the ego receiver selects and requests mode s at frame t . We model the effective frame F1 under

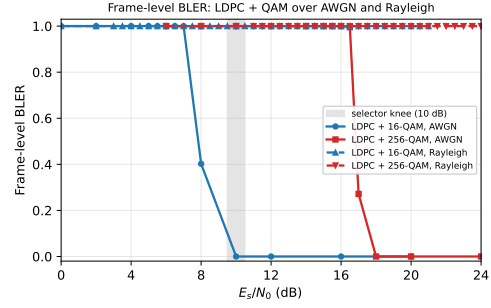


Fig. 2. Block-error rate of the feature-level message under LDPC 1/2 with 16-QAM and 256-QAM, over AWGN (solid) and Rayleigh (dashed) channels. The 16-QAM AWGN cliff (an 8–12 dB transition, onset 8.0 dB) defines the selector knee (shaded); the 256-QAM cliff sits far to its right (onset 16.5 dB), so by the time 256-QAM is reliable the 16-QAM block is already error-free. Under Rayleigh fading both curves degrade so slowly that, under the all-or-nothing frame model evaluated here, feature-level transmission never becomes reliable in 0–20 dB, explaining the selector's conservative L -only behaviour.

channel corruption as

$$F_t^L = F_t^{L, \text{clean}}, \quad (5)$$

$$F_t^{C_q} = (1 - \text{BLER}_q(\gamma_t, \text{ch}_t)) F_t^{C_q, \text{clean}} + \text{BLER}_q(\gamma_t, \text{ch}_t) F_t^{\text{ego}}, \quad q \in \{16, 256\}, \quad (6)$$

where $\text{ch}_t \in \{\text{AWGN}, \text{Rayleigh}\}$, $F_t^{L, \text{clean}}$ denotes the frame F1 under a perfect channel, and F_t^{ego} is the frame F1 of the ego vehicle's own single-agent (pre-fusion) detection. Equation (6) is an ego-fallback block-loss model: with probability BLER_q the feature-level message is lost and the receiver reverts to its ego-only output F_t^{ego} ; with probability $1 - \text{BLER}_q$ it is delivered intact and contributes its clean F1. Because only one message is requested per frame, a lost feature block leaves no collaborator information, so the fallback is the ego-only detection rather than a zero-utility state; both this frame-F1 model and the true end-to-end AP evaluation of Section VI-D realise the same per-frame fallback.

E. Communication Cost

Let B_s denote the per-frame payload in channel-use-equivalent mega-symbols (Msym) for mode s , accounting for the spectral efficiency of the modulation:

$$B_L = 0.024, \quad B_F = 3.96/4 \approx 0.99, \quad B_{C_{256}} = 3.96/8 \approx 0.495, \quad (7)$$

all in Msym/frame, where $3.96 = 1.98/(1/2)$ is the coded-bit count produced by rate-1/2 LDPC from the 1.98-Mbit feature payload, and the divisors 4 and 8 are the bits-per-symbol of 16-QAM and 256-QAM, respectively. The object-level message is sent as a protected low-rate message at approximately 1 bit per channel use (rate-1/2 QPSK), so its 0.024 Mbit of information occupies ≈ 0.024 Msym. The average communication cost over T evaluated frames is

$$\bar{B} = \frac{1}{T} \sum_{t=1}^T B_{s_t} = \sum_{s \in S} \rho_s B_s, \quad (8)$$

where ρ_s is the fraction of frames assigned to mode s . This equation highlights the role of semantic granularity selection: reducing the frequency of high-payload feature-level transmission directly reduces the average communication cost.

F. Task-Communication Objective

The goal of channel-aware semantic granularity selection is to choose a policy that balances detection utility and communication cost. Let z_t denote the online task cue vector extracted from the ego-side perception stream, and let γ_t denote the estimated channel quality. The selector is defined as

$$s_t = g(z_t, \gamma_t, c_t), \quad (9)$$

where $g(\cdot)$ maps the current task state and channel state to one of the message modes in $\mathcal{S}$.

In this paper we instantiate the policy via a channel-aware utility argmax,

$$s_t^* = \arg \max_{s \in \mathcal{S}} F_t^s, \quad (10)$$

which is the $\lambda = 0$ special case of the payload-penalised objective introduced in Eq. (13) and adopted as the formal training target. Of the three deployed budgets only $B_{\max} = 0.30$ freezes at $\lambda^* = 0$ and is therefore described by this form; the other two carry a strictly positive penalty.

G. Derivation: Constrained Optimisation and Rate-Distortion View

Equation (10) is the Lagrangian-relaxed optimum of the constrained perception-communication problem

$$\max_g \mathbb{E}[F_t^{s_t}] \quad \text{s.t.} \quad \mathbb{E}[B_{s_t}] \leq \bar{B}_{\max}, \quad (11)$$

where $\bar{B}_{\max}$ is the prespecified average communication budget—a bound on the mean per-frame payload over a split, not a per-frame cap. Introducing the Lagrange multiplier $\lambda \geq 0$, the relaxed problem is

$$\max_g \mathbb{E}[F_t^{s_t}] - \lambda \mathbb{E}[B_{s_t}], \quad (12)$$

whose pointwise optimum is

$$s_t^*(\lambda) = \arg \max_{s \in \mathcal{S}} (F_t^s - \lambda B_s). \quad (13)$$

The learned selector is trained against Eq. (13), not against Eq. (10): the payload penalty λ is a selected quantity, frozen per budget by the walk of Section V at $\lambda^* = 0.05, 0.02$ and 0.00 for $B_{\max} = 0.10, 0.20$ and 0.30 respectively. Equation (10) is thus the $\lambda = 0$ member of the family and describes only the $B_{\max} = 0.30$ operating point; the two tighter budgets are trained with a strictly positive penalty, and the constrained problem itself is addressed by sweeping λ over the family (the supplementary material).

In rate-distortion terms, Eq. (6) is a channel-induced distortion scaling with $\text{BLER}_q(\gamma)$: mode C_q delivers rate $R_{C_q} = 3.96/\log_2 M_q$ Msym (0.99 and 0.495 for C_{16} and C_{256}) at survival probability $1 - \text{BLER}_q(\gamma)$, while L

delivers a much lower rate at effectively unit survival. The selector therefore trades rate against rate-conditional distortion as a function of channel state—joint source-channel coding structure, expressed at the granularity-action level rather than the symbol level.

This formulation leads to a key design principle: the most informative message is not always the most useful message under communication constraints. A feature-level message may provide richer semantics, but if the channel is poor, its realized task utility may be lower than that of a compact object-level message. Therefore, the selector must consider both the task state and the channel state before deciding the semantic granularity of the transmitted message.

IV. Proposed Method

A. Overview

The proposed CA-TOSG framework is designed to select the semantic granularity of V2V communication according to both the perception task state and the channel condition. Instead of transmitting a fixed representation for all frames, the ego vehicle chooses one message mode from $\mathcal{S} = \{E, L, F\}$ —ego-only, object-level, or feature-level—before any high-payload transmission, signals the choice to the collaborator via the 2-bit CAM-embedded request introduced in Section III, and the collaborator complies.

Fig. 1 illustrates the overall pipeline. The ego vehicle first extracts a local BEV feature and local perception outputs using a shared PointPillars backbone [25]. Based on these outputs, it constructs an online task cue vector z_t (Section IV-B). Meanwhile, the current channel quality is represented by the estimated SNR $\hat{\gamma}_t$ and the channel type c_t , both produced by the ego's receiver-side link adaptation module already present in 802.11bd / 5G NR sidelink stacks. A lightweight selector takes $(z_t, \hat{\gamma}_t, c_t)$ as input and predicts the communication mode s_t . The collaborator transmits the requested message: if $s_t = L$, compact object-level detections; if $s_t = F$, a compressed feature-level message under 16-QAM; if $s_t = E$ the ego issues no request at all and operates on its own perception. At the ego, the received message is fused with $F_{\text{ego},t}$ to produce the final detection result.

The key idea is that semantic richness and task utility are not equivalent under channel constraints. Feature-level messages contain richer intermediate representations, but they are more sensitive to channel degradation and require higher communication resources. Object-level messages are less expressive but compact and robust. Therefore, the selector should activate feature-level communication only when the channel condition and task state indicate that the additional semantic information is likely to improve detection.

B. Ego-Side Online Task Cues

The first input to the selector is the ego-side task cue vector $z_t \in \mathbb{R}^{21}$. Five are representative of the whole: the ego's own detected-object count, the LiDAR point count,

the near-range point density 0–20 m, the mean point range, and the front-sector far-range count beyond 30 m; the full definition list of all 21 is in the supplementary material. Every cue is computed from the ego vehicle's own LiDAR observation and the ego's own local detection stream; no information from the collaborator's perception stack is required at decision time. This is a direct consequence of the receiver-driven signalling architecture introduced in Section III and removes the causality issue that would arise if the collaborator had to estimate ego-side state without feedback. Pre-transmission, the ego runs the selector, signals the chosen mode s_t to the collaborator via the 2-bit CAM request, and only then does the collaborator transmit $m_t^{s_t}$.

These cues collectively describe local sensing quality, scene geometry, and detection load, all of which influence whether the current frame benefits from richer feature-level information from a collaborator. Confidence-related and temporal-stability cues can be added straightforwardly when intermediate detection outputs are accessible to the cue extractor; the framework does not depend on any particular cue choice.

C. Channel Quality Cue

The second input is the channel quality, represented by the estimated SNR $\hat{\gamma}_t \in \mathbb{R}$ in dB and a binary channel-type indicator $c_t \in \{0, 1\}$ encoding AWGN versus Rayleigh fading. In a deployed V2X stack, $\hat{\gamma}_t$ is obtained from pilot-symbol SNR estimation already produced by 802.11bd or 5G NR sidelink receivers [3], and c_t can be inferred from link-layer multipath indicators such as RMS delay spread or Doppler estimates.

The role of $(\hat{\gamma}_t, c_t)$ is fundamentally different from the perception cues. Task cues estimate whether the current frame needs richer semantic information; the channel cues estimate whether such information can be transmitted reliably. A difficult frame does not necessarily imply that feature-level communication should be used; if the channel is poor, transmitting a corrupted feature-level message may not improve detection and may waste bandwidth. Conversely, when the channel is strong, feature-level communication may become worthwhile for frames where object-level messages are insufficient. The selector must therefore consider the joint state $(z_t, \hat{\gamma}_t, c_t)$, which is the foundation of CA-TOSG.

D. Message Branches

1) **Object-Level Branch:** The object-level branch transmits compact detection results $m_t^L = \{(b_k, c_k, p_k)\}_{k=1}^{N_t}$. Its average payload is approximately $B_L = 0.024$ Msym/frame, allowing it to be transmitted with strong channel coding and treated as channel-invariant in the evaluated SNR range.

2) **Feature-Level Branches:** The feature-level branches transmit compressed BEV representations $m_t^{C_q} = \phi_q(F_{j,t})$, where $\phi_q(\cdot)$ uses q -QAM modulation under rate-1/2 LDPC coding. The C_{16} mode is more reliable but

less spectrally efficient (channel-use payload $B_{C_{16}} \approx 0.99$ Msym/frame), while the C_{256} mode is more spectrally efficient but more channel-sensitive (channel-use payload $B_{C_{256}} \approx 0.495$ Msym/frame).

E. Channel-Aware Semantic Granularity Selector

The selector implements the policy $g(z_t, \hat{\gamma}_t, c_t)$ of Eq. (9) as a Random Forest classifier. The three deployed selectors are the ones the pre-registered budget walk froze (FROZEN_MANIFEST.json): $N_T = 400$ trees and `max_features=sqrt` throughout, minimum samples per leaf of 2, a depth bound of 10 at $B_{\max} = 0.10$ and 0.20 and unbounded depth at $B_{\max} = 0.30$, and `class_weight=None` at all three budgets — the walk selected unweighted candidates, so no class re-weighting is applied to the oracle-label imbalance. We choose Random Forest for three practical reasons. First, the selector should be training-free with respect to the perception backbone and detection head: it should be deployable on top of any pretrained cooperative perception stack without retraining the perception model. Random Forest fits this constraint because it operates purely on tabular cues. Second, Random Forest provides interpretable feature importance (Table IV), which is valuable for safety certification of vehicular systems. Third, its per-frame inference cost on a single CPU core is 52.1 ± 5.6 ms (P95 = 58.3 ms), measured over 1,000 batch-1 trials per frozen selector and quoted for the slowest of the three (the supplementary material); this fits the 100 ms budget of a 10 Hz LiDAR cycle without requiring a GPU, dedicated accelerator, or framework-level optimisation. We emphasise that the contribution is not the Random Forest itself: it is an interpretable implementation of the channel-aware granularity policy. The SNR-threshold and three-scalar hand-rule baselines achieve comparable F1, but require higher communication payload at the primary operating point (Section VI-G).

The selector is trained to imitate the channel-aware oracle defined in Eq. (10). The oracle uses ground-truth detection outputs and is therefore not deployable; it is used only to generate training labels and as an upper-bound reference. Before taking the argmax in Eq. (10), the oracle applies a feasibility mask: any mode whose frame-level block-error rate exceeds 0.999 at the operating (γ_t, c_t) grid point is removed from $\mathcal{S}$, so that neither the oracle labels nor the deployed selector request a message the channel almost surely cannot deliver; when a requested feature block is nonetheless lost, the pipeline reverts to the ego-only output rather than a zero-utility state (Eq. (6)). At deployment, the selector uses only the online cues z_t and the channel state $(\hat{\gamma}_t, c_t)$:

$$s_t = g(z_t, \hat{\gamma}_t, c_t). \quad (14)$$

This design gives the selector an interpretable behavior. At low SNR, feature-level messages are likely to be unreliable, so the selector tends to choose L . At moderate SNR, C_{16} may become useful because it provides richer features with better reliability than C_{256} . Under fading

channels, the selector remains more conservative because the average SNR does not fully eliminate deep fading risk. We will see in Section V that the empirical behavior of the trained selector follows this pattern.

F. Receiver-Side Fusion and Detection

After transmission, the ego vehicle receives the selected message $m_t^{s_t}$. The receiver fuses this message with the local ego feature:

$$\hat{Y}_t = D(\Psi(F_{\text{ego},t}, m_t^{s_t})), \quad (15)$$

where $\Psi(\cdot)$ denotes the cooperative fusion module and $D(\cdot)$ is the detection head. The fusion operation depends on the selected message type: if $s_t = L$, the receiver performs object-level fusion using the received detections; if $s_t = F$, the receiver decodes the feature-level representation and performs intermediate fusion.

G. Discussion of Design Advantages

The proposed design has three main advantages. First, it avoids unnecessary feature-level communication in easy frames or poor channels, reducing the average payload. Second, it prevents unreliable feature messages from degrading perception under low SNR or fading. Third, it remains compatible with existing feature-level communication modules. For example, Where2comm [5], JSCC-based codecs [6], or other semantic communication methods can be used inside the C_{16} or C_{256} branches, while CA-TOSG decides when these branches should be activated. This separation between semantic granularity selection and feature-level transmission is important for practical vehicular systems. It allows the communication policy to adapt at the frame level without redesigning the entire perception backbone or communication codec.

V. Experimental Setup

A. Dataset and Implementation

We evaluate CA-TOSG on the OPV2V dataset [1] using the OpenCOOD codebase. OPV2V is a large-scale simulated V2V cooperative perception dataset that provides multi-agent LiDAR observations, vehicle poses, and 3D object annotations under connected-vehicle settings. We use the validate split, which contains $T = 1,980$ frames, for all primary results, and additionally verify generalisation on the held-out OPV2V test split ($T = 2,170$ frames, scene-disjoint from validate) and the Culver-City split ($T = 550$ frames, a real-road-layout domain shift) in Section VI-E, applying the validate-trained selector without retraining. The perception model is a PointPillars BEV backbone [25] with detection range $x \in [-140.8, 140.8]$ m and $y \in [-38.4, 38.4]$ m. The backbone and detection head are taken from the public OpenCOOD pretrained checkpoints and are frozen throughout this work; CA-TOSG adds no learnable parameters to the perception pipeline. All methods share the same backbone and detection head to ensure a fair comparison.

B. Message Construction and Payload Accounting

Both payloads are made explicit rather than left implicit: the object-level payload is derived from first principles, while the feature-level source budget is a declared convention to which, as shown below, all conclusions are insensitive. The object-level message L carries the collaborator's detected objects; on the OPV2V validate split a frame contains on average 27 detected objects, and encoding each as an ETSI-CPM-style perceived-object container (3D box, dimensions, class, confidence and position covariance, ≈ 110 bytes) gives $B_L \approx 27 \times 110 \times 8 \approx 0.024$ Mbit/frame—a deliberately conservative figure that, if anything, overstates the object-level cost. The feature-level message encodes the transmitted BEV feature tensor of size $256 \times 48 \times 176 \approx 2.16 \times 10^6$ elements: the 281.6×76.8 m detection range at 0.4 m voxels yields a 704×192 pillar grid, down-sampled by 4 to 48×176 with 256 channels. We adopt a fixed source budget of $B_C \approx 1.98$ Mbit/frame for the feature message, i.e. ≈ 0.92 bit per element of this 2.16×10^6 -element tensor; because every comparison in this paper is ratio-like (channel-use fractions and payload-F1 orderings), the conclusions are insensitive to this constant—re-anchoring to 2.16 Mbit would rescale the feature cost of all policies equally. The feature-level message is therefore $\approx 82\times$ the object-level payload. Both feature modes C_{16} and C_{256} carry this same 1.98 Mbit perception payload but require different numbers of channel uses; applying the rate-1/2 coded-bit conversion of Eq. (7) gives $B_{C_{16}} \approx 0.99$ Msym/frame for 16-QAM and $B_{C_{256}} \approx 0.495$ Msym/frame for 256-QAM. All payload comparisons in this paper use this channel-use-equivalent definition.

C. Channel Settings

We sweep the estimated SNR over $\gamma \in \{0, 2, 4, \dots, 20\}$ dB, giving 11 SNR points per channel. Training and the headline evaluation use two channel models, AWGN and Rayleigh fading; the frequency-selective OFDM link of Section III-C (TDL, $\tau_{\text{rms}} = 46.2$ ns, empirical diversity order 2.01) is additionally used in the codec-comparison analysis of the supplementary material. The selector's training substrate is the full deterministic grid—every one of the 1,980 frames crossed with all 11 SNR points and both channel types (43,560 rows)—rather than one sampled (γ, ch) per frame; the 200-realisation Monte-Carlo draw is used for evaluation, not for building labels. The BLER functions $\text{BLER}_q(\gamma, \text{ch})$ are tabulated from a separate LDPC + QAM simulation; we instantiate this transport with the 5G NR LDPC (Sionna) under 3GPP TR 37.885 Urban NLOSv [4]. This transport configuration—rate-1/2 LDPC with 16-/256-QAM, compared at the same coded-bit count—matches the conventional digital baseline of SComCP [7]. Rayleigh is simulated directly at link level in the same Sionna chain as AWGN—per-codeword flat block fading with $|h|^2 \sim \text{Exp}(1)$ and perfect receiver CSI, so each block sees its own effective E_s/N_0 —

and is not obtained by averaging the AWGN table over an instantaneous-SNR distribution. Frame-level BLER follows from the measured codeword BLER as $1 - (1 - \text{BLER}_{\text{cw}})^{N_{\text{cw}}}$ with $N_{\text{cw}} = 3,960$ codewords per feature frame.

D. Selector Training

The three deployed selectors are the ones the pre-registered budget walk froze: scene-level 9-fold LOSO on validate, a frozen candidate walk under the hard payload constraint, and per-budget $\lambda^* = 0.05/0.02/0.00$ with $\tau^* = 18/12/8$ dB for $B_{\text{max}} = 0.10/0.20/0.30$ (FROZEN_MANIFEST.json). The candidate set, the walk order and the tie-break are in the supplementary material.

E. Baselines

We compare CA-TOSG with the following baselines:

Fixed L . The sender always transmits the object-level message. This is treated as channel-invariant in the evaluated SNR range.

Fixed F . The sender always transmits the feature-level message under 16-QAM coding.

Fixed C_{256} . The sender always transmits the feature-level message under 256-QAM coding.

SNR-threshold rule (τ^*). A one-line channel rule—request F under AWGN when $\hat{\gamma}_t > \tau^*$, otherwise L —with τ^* tuned on validate for each budget. Reported both at its nominal tuning and at the strictly budget-matched τ_{feasible} .

Two- and three-scalar hand rules. Threshold policies over the same committed cues, fitted through the identical scene-level LOSO walk and budget constraint as the selector: a two-scalar rule (difficulty gate plus link-reliability gate) and, after the two-scalar form proved unable to meet the two tighter budgets, a three-scalar amendment whose feature gate is a conjunction of the two. They are the strongest hand-designed comparators in this paper and are analysed in Section VI-G.

Channel-aware oracle. For each frame, the oracle selects the message mode that maximises Eq. (6) given the realised SNR and channel; it is not deployable but serves as an upper-bound reference.

ImportanceMapJSCC (exploratory; the supplementary material). The learned semantic communication baseline of [6], evaluated under the same perception backbone and the same AWGN/Rayleigh channel models. It is a prior-protocol arm: it was not re-evaluated under the frozen single-collaborator protocol and therefore appears in no mainline table or figure—every result involving it is confined to the supplementary material and is exploratory. This is the strong feature-level communication baseline. It is our reproduction of the published method under our perception backbone, dataset split and channel tables—no code, checkpoint or trained selector from the original authors is used—so it should be read as a faithful re-implementation rather than as the authors’ own system.

LDPC + 16/256-QAM (separate coding). Standard separate source-channel coding baselines using rate-1/2 LDPC with 16-QAM and 256-QAM modulation [3]. These represent traditional digital communication of the same feature-level payload as ImportanceMapJSCC.

Perfect-channel upper bound. Detection performance under ideal (lossless) feature-level transmission, included as a saturating ceiling for the AP curves.

F. Evaluation Metrics

We report AP@0.5 and AP@0.7 for end-to-end detection performance, mean realised frame F1 for fine-grained frame-level behaviour, and average payload in channel-use-equivalent Msym/frame for communication cost. We also report the selection ratios $\rho_s = (1/T) \sum_t \mathbf{1}(s_t = s)$ to expose how the selector’s policy varies with channel quality.

VI. Results and Analysis

A. Headline Results

Feature-level cooperation pays only when the channel can deliver it, so a channel-averaged number understates the effect. We therefore report true end-to-end AP in the two regimes that define the policy: the fallback regime (fading or low-SNR AWGN), where CA-TOSG holds L at 0.024 Msym/frame, and the feature-active regime (AWGN $\hat{\gamma}_t \geq 10$ dB, the measured ρ_F knee of Section VI-D, not a stipulated threshold). AP is computed by concatenating cached per-frame predictions according to the selector’s choice and scoring against OPV2V ground truth (full protocol in Section VI-D); the per-SNR breakdown and the Rayleigh curves are in the committed product `true_e2e_ap_by_snr.csv`.

Three observations follow. (i) The realised true end-to-end AP@0.5 of the frozen selectors, reported descriptively for $B_{\text{max}} = 0.10/0.20/0.30$, is 0.7887/0.7926/0.7936 on validate, 0.8697/0.8742/0.8742 on test and 0.7299/0.7347/0.7504 on Culver-City, against fixed object-level references of 0.7819, 0.8691 and 0.7299. Every split is read the same way. What any granularity policy can contest is the headroom between the perfect-channel feature ceiling and fixed L : $0.8369 - 0.7819 = 0.0550$ AP on validate, $0.8931 - 0.8691 = 0.0240$ on the scene-disjoint test split and $0.8269 - 0.7299 = 0.0970$ on Culver-City. The selector’s realised share, $(\text{AP}_{\text{CA-TOSG}} - \text{AP}_{\text{Fixed } L})/\text{headroom}$, is 12.4/19.5/21.3% on validate, 2.5/21.2/21.2% on test and 0.0/4.9/21.1% on Culver-City—at most about a fifth, so what it reliably delivers is the communication saving, not an AP gain, and these figures are descriptive. Under the single-collaborator convention the L - F gap is wider than it was, because box-level fusion of one neighbour discards more than feature-level fusion of the same neighbour; the retired full-collaborator accounting showed a test headroom an order of magnitude smaller, an artefact of fusing every collaborator into both branches. (ii) Under fading or low SNR the selector holds L and loses no

TABLE II

Headline results: true end-to-end AP of CA-TOSG across three OPV2V splits, in the fading/low-SNR fallback regime (selector holds L) and the AWGN $\hat{\gamma}_t \geq 10$ dB feature-active regime (selector requests F), the boundary being the measured ρ_F knee of Section VI-D, averaged over the six grid points at and above it. When the channel can carry features, CA-TOSG lifts AP on all three splits (+0.047 validate, +0.020 test, +0.018 Culver-City at IoU 0.5); when it cannot, the selector safely retains the L operating point. Payload is per-frame channel use (Msym, at rate-1/2); channel-averaged CA-TOSG payload is 0.068–0.151 Msym/frame across splits, i.e. 7–15% of Fixed F (0.990).

Split	Regime	AP@0.5	AP@0.7	Payload
Validate (1,980)	Fallback (L)	0.7818	0.7038	0.029
	AWGN ≥ 10 dB	0.8290	0.7296	0.483
Test (2,170)	Fallback (L)	0.8691	0.8027	0.024
	AWGN ≥ 10 dB	0.8892	0.8177	0.453
Culver-City (550)	Fallback (L)	0.7299	0.6463	0.024
	AWGN ≥ 10 dB	0.7475	0.6553	0.180

AP against object-level communication, while the fixed feature-level policies C_{16} and C_{256} are strictly dominated: 21–41 $\times$ more channel use, below Fixed L in realised AP under the cliff (Fig. 3). (iii) Channel-averaged, the selector spends 0.08102–0.20361 Msym/frame on validate, 0.03680–0.21196 on test and 0.02437–0.18226 on Culver-City—8.2–20.6%, 3.7–21.4% and 2.5–18.4% of Fixed F —because the feature-active regime occupies only the high-SNR AWGN tail.

CA-TOSG is not designed to raise channel-averaged AP above Fixed L on every frame—on many frames object-level communication already suffices and the average dilutes the gain—but to activate feature-level communication only when the expected task gain justifies the payload and channel risk; the payload–F1 frontier (the supplementary material), the difficulty stratification (the supplementary material) and the threshold comparison (Section VI-F) localise where.

Because a one-line SNR-threshold rule is the natural challenger to any learned selector, we place it in the headline rather than an appendix. Table III reports the channel-averaged frame F1 and per-frame channel use of Fixed L , a re-tuned SNR-threshold rule, the learned selector, and the oracle over 200 random SNR/channel draws on the frozen test split. The comparison is budget-indexed, and it does not run one way. At $B_{\max} = 0.10$ the selector’s realised F1 is now marginally below that of a threshold tuned on validate for the same target budget, by -0.00099 (frame-level paired bootstrap 95% CI $[-0.00105, -0.00093]$) — the direction reversed under the corrected convention, and it is reported as it came out; this is a secondary comparison and we report the interval only, with no decision attached. At $B_{\max} = 0.20$ and 0.30 the nominal threshold rule attains the higher F1—0.89701 and 0.89900 against 0.89691 and 0.89783—while transmitting 1.53 $\times$ and 1.47 $\times$ the channel use, and doing so from outside the budget; against the budget-matched τ_{feasible} the F1 ordering at $B_{\max} = 0.20$ reverses in the selector’s favour. The claim that survives is the pre-registered one, on the single confirmatory comparison (test at $B_{\max} = 0.20$): the selector’s F1 is lower than the nominal threshold rule’s by only ≈ 0.0001 (95% CI

upper bound -0.00002 , still entirely below zero) yet far within the pre-registered non-inferiority margin of 0.005, while reducing communication payload by 34.8% against a nominal threshold that is itself over budget ($0.2168 > 0.20$ Msym), or 26.6% against the budget-matched τ_{feasible} . Frames within a scene are not independent; resampling the same comparison at the level of the 16 test scenes widens both intervals without changing the ruling, and both are reported together in Table VI. On an LDPC + QAM cliff, in other words, channel state carries most of the accuracy decision — most, not all, and only on this transport — and what the ego-side cues buy is bandwidth, not peak accuracy. We report this rather than hide it: it is the central diagnostic finding of the paper, and it recurs as the codec-dependent currency of the task cues (the supplementary material).

A qualitative example of such a frame—object-level F1 0.67 against feature-level 0.95 on the same frame, with the object-level branch producing false positives away from any ground-truth vehicle—is recovered by the repository’s figure product (plot_qualitative_bev.py); the selector’s job is to identify such frames and request F only when the channel permits.

B. Detection Performance versus SNR

Fig. 3 reports the frozen selectors’ realised effective F1 against SNR under both channels, beside Fixed L , Fixed F , the perfect-channel feature ceiling and the feasibility-masked oracle. Every curve is read from the same frozen grid product (frozen_curves.csv); the codec-comparison baselines (importance-map JSCC and the LDPC + 16/256-QAM separate-coding arms) are a prior-protocol arm and are reported in the supplementary material rather than mixed into this figure.

Under AWGN at $B_{\max} = 0.20$ the selector sits on the Fixed L line (0.8554 on validate) for SNR ≤ 8 dB and steps to 0.8875 at 10 dB, where it stays flat to 20 dB. Two things about that step matter. It is a policy step, not a detection-quality step: it happens exactly where the 16-QAM frame-error cliff clears (Section III-C), and above it the curve is flat—there is no further AP to be had from a better channel. And it does not reach the

TABLE III

Channel-averaged comparison on the frozen OPV2V test split (selector trained on validate, not retrained; $T = 2,170$ frames), read from the frozen replay: F1 and channel use are the means over the same 200 paired CSI samplings ($\text{SNR} \sim \mathcal{U}[0, 20]$ dB, 50/50 AWGN/Rayleigh) for every row. The comparison is budget-indexed: each frozen selector is shown against the SNR-threshold rule tuned to the same budget, and it does not run one way. Share is channel use as a fraction of Fixed F (0.990 Msym).

Policy	Mean F1	Channel use (Msym)	Share of Fixed F
Fixed L (object-level)	0.8910	0.024	2.4%
Fixed F (LDPC+16-QAM)	0.8483	0.990	100.0%
Fixed C_{256} (LDPC+256-QAM)	0.8255	0.495	50.0%
Masked oracle (upper bound)	0.9056	0.175	17.7%
$B_{\max} = 0.10$			
SNR-threshold ($\tau^*=18$)	0.89247	0.0724	7.3%
CA-TOSG (RF, frozen)	0.89148	0.0368	3.7%
$B_{\max} = 0.20$			
SNR-threshold ($\tau^*=12$)	0.89701	0.2168	21.9%
CA-TOSG (RF, frozen)	0.89691	0.1414	14.3%
$B_{\max} = 0.30$			
SNR-threshold ($\tau^*=8$)	0.89900	0.3125	31.6%
CA-TOSG (RF, frozen)	0.89783	0.2120	21.4%

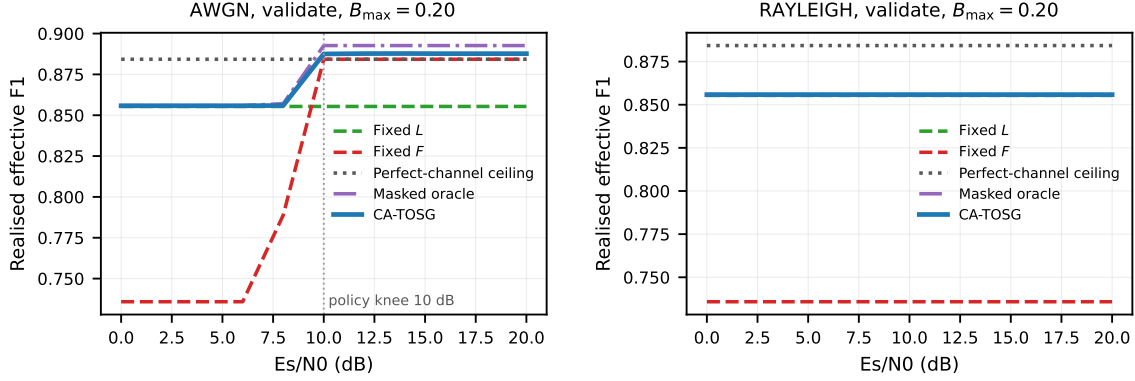


Fig. 3. Realised effective F1 versus SNR under AWGN (left) and Rayleigh (right) on OPV2V validate at $B_{\max} = 0.20$, every curve from the frozen grid. CA-TOSG holds L below the cliff—its own realised F1 there is 0.8558, marginally above the Fixed- L line at 0.8554 because of the E/L mix—and steps to 0.8875 at the measured 10 dB knee, reaching 0.8877 by 20 dB; the perfect-channel feature ceiling is 0.8843 and the feasibility-masked oracle reaches 0.8927, so the step recovers part of the headroom rather than all of it. Under Rayleigh the selector stays on Fixed L throughout, flat at 0.8558 across the whole range. The codec-comparison baselines are a prior-protocol arm and are in the supplementary material.

ceiling: the perfect-channel feature reference is 0.8843 on validate and the masked oracle reaches 0.8927, so the frozen selector recovers part of the available headroom, not essentially all of it. Under Rayleigh the selector remains on Fixed L across the entire SNR range; the deep-fade frame BLER never falls low enough for the feature branch to be worth requesting, even at 20 dB mean SNR. No fixed feature-level policy beats object-level communication on a channel-averaged basis—the low-SNR cliff outweighs the high-SNR gain—which is precisely the per-frame gating opportunity CA-TOSG exploits by holding L below the threshold and activating F above it.

Payload versus SNR. On the same frozen grid at $B_{\max} = 0.20$, CA-TOSG holds the L -payload (0.0237 Msym/frame on validate—the E/L mix, measured at AWGN 0 dB and identical at every Rayleigh SNR) at every SNR under Rayleigh and for AWGN SNR ≤ 8 dB, then steps once at 10 dB to 0.4795 Msym and stays there—about 48% of the Fixed- F payload, not a near- F payload: even with the

TABLE IV

Top-7 features by Gini importance for the deployed CA-TOSG selector (trained on OPV2V validate, $T = 1,980$ frames). The two channel-side features account for 61.7% of total importance; the 21 perception-side cues carry the remaining 38.3%. Read directly from the frozen model, results/main/feature_importance_frozen.csv.

Feature	Gini importance
channel_is_rayleigh	0.342
est_snr_db	0.275
pcd_mean_range	0.030
pcd_left_points	0.028
ego_num_objects	0.025
pcd_std_range	0.023
pcd_front_far_30m	0.021
$\hat{\gamma}_t + c_t$ subtotal	0.617

channel fully open the frozen selector requests the feature message on only about a quarter of frames. The fixed-mode baselines are flat at their constant payload and give the worst-case bandwidth reference.

C. Ablation: Effect of Channel-State Features

To isolate the contribution of the two channel-state features, we train selector variants under identical RF hyperparameters over the feature subsets listed in the supplementary material: perception cues only (21 LiDAR cues); the perception cues plus the estimated SNR $\hat{\gamma}_t$; the full configuration adding the channel-type indicator c_t (the deployed selector); and, for reference, channel state alone ($\hat{\gamma}_t, c_t$) and a hand-tuned SNR threshold. The full comparison table is in the supplementary material.

Two findings stand out. First, neither half of the input is sufficient on its own, and the failure mode is the same in both directions: run through the identical pipeline under $B_{\max} = 0.20$, the channel-only variant (2 features) and the cues-only variant (21 features) both collapse to always transmitting L —feature-request rate $\rho_F = 0.000$, payload pinned at B_L , realised F1 at the fixed- L floor 0.8909—while the full selector reaches 0.8969 at 0.141 Msym. Cues without a channel signal cannot tell whether a feature request would survive the link; channel state without cues cannot tell whether it would be worth sending. Only the combination yields a policy that sits between the two extremes. The collapse has a single mechanism, and it explains the same behaviour wherever it appears in this paper. The committed frame BLER is effectively binary in the channel state—1.0 for Rayleigh at every tabulated SNR, and for AWGN below 8 dB, against 0.0 at and above 10 dB—so a policy whose feature decision depends on the channel alone can realise only four mean payloads, 0.024, ≈ 0.27 , ≈ 0.33 and 0.99 Msym, and the two tighter budgets fall between those rungs. That is why the channel-only variant pins itself to B_L at $B_{\max} = 0.10$ and 0.20 and only activates F at 0.30; why the pre-registered two-scalar hand rule degenerates to Fixed L at the same two budgets (Section VI-G); and why the nominal SNR threshold lands over budget at $B_{\max} = 0.20$ rather than on it. Reaching an intermediate budget requires deciding which frames deserve F among those whose link can carry one, which is a per-frame content decision and is exactly what the perception cues supply. This does not contradict the sufficiency of the SNR threshold in Section VI-F: SNR is decisive as the threshold signal that gates the feature request under AWGN, but as an additional RF feature atop the perception cues it adds a noisy continuous dimension without an F1 gain—the two roles are distinct. Second, what the perception cues buy is deployed channel use rather than accuracy, and the effect is budget-dependent. At $B_{\max} = 0.10$ and 0.20 the channel-only variant has no graded policy to offer at all ($\rho_F = 0$, payload pinned at B_L). At $B_{\max} = 0.30$ it does activate the feature action ($\rho_F = 0.274$), but under the corrected convention it is now beaten on both axes at once: on test it reaches 0.89529 against the full selector’s 0.89783 (-0.0025 F1) while spending 0.287 Msym against 0.212— $1.36\times$ the channel use for less accuracy. Under the retired accounting this variant bought $+0.0021$ F1 for that extra spend; the correction removes even that.

Four independently constructed variants collapse in the same way on this substrate — channel-state-only, cues-only, a contextual-bandit comparator retrained on the corrected grid (which selects one λ for all three budgets and settles at a near- L operating point, 0.0294 Msym), and the SECOND-backbone selector of the supplementary material — and only the full-feature imitation selector produces three distinct, budget-indexed operating points. The comparison is about which construction yields a graded policy, not about one method outperforming another. Under the corrected convention the cues do add accuracy on this axis ($+0.0090$ over channel state alone); what they mainly buy is still that comparable accuracy can be bought more cheaply.

D. True End-to-end AP Verification

The realised frame F1 reported so far is computed under the analytical block-loss model of Eq. (6), where a lost feature block falls back to the ego vehicle’s own pre-fusion detection. To confirm that this analytical F1 proxy translates to actual end-to-end detection AP under a deployed pipeline, we run a true end-to-end inference in which: (i) the deployed selector picks s_t per frame from $(\hat{\gamma}_t, c_t)$; (ii) cached per-frame predictions from the OPV2V late-fusion checkpoint (for L) and the attentive-compression checkpoint (for C_q) are concatenated according to the selector’s choice; (iii) each C_q frame is dropped with probability $\text{BLER}_q(\gamma_t, c_t)$ and falls back to the ego-only (pre-fusion) prediction; (iv) the resulting box set is scored against ground-truth boxes with the OpenCOOD VOC-style AP computation at IoU thresholds $\{0.3, 0.5, 0.7\}$, averaged over the 200 paired stochastic channel realisations of the frozen replay. The per-SNR breakdown is the committed product `true_e2e_ap_by_snr.csv`.

Two observations support the central claim. First, the boundary of the feature-active regime is measured, not stipulated: reading the frozen selector’s feature-request rate off the pre-registered SNR grid, ρ_F is zero at every AWGN point up to 8 dB and steps to its plateau at 10 dB on all three splits—0.472 on validate, 0.433 on test and 0.160 on Culver-City—which is exactly where the 16-QAM frame-error cliff clears (Section III-C).³ At that boundary the true end-to-end AP on validate is 0.8287 at IoU 0.5 and 0.7292 at IoU 0.7, and it is flat above it. The selector reaches this while paying 0.300 Msym/frame averaged over the feature-active regime instead of the 0.990 of always-on Fixed F —a 70% saving—though on validate it recovers only part of the perfect-channel reference of the underlying attentive-compression model (0.8369 on validate, Section VI-A) rather than essentially all of it, as an earlier version of this section claimed. Second, the Rayleigh AP curve is flat across the

³Validate shows a 0.4% toe at 8 dB, the single grid point at which the frame BLER is neither ≈ 0 nor ≈ 1 (0.402); test and Culver-City are exactly zero there. We therefore take 10 dB as the regime boundary on all three splits and report the toe rather than smoothing it away. Earlier versions of this section stipulated ≥ 14 dB; that threshold was not derived from any measurement and is withdrawn.

full SNR range and matches Fixed L , confirming that the selector’s conservative fading-regime behaviour, identified in Section VI-A, is not a frame-F1 artefact but a genuine end-to-end property.

E. Generalisation to OPV2V Test and Culver-City Splits

The results above are computed on validate. We re-run the entire pipeline—per-frame inference, cues, oracle labelling and true end-to-end AP scoring—on two held-out splits: the scene-disjoint test split ($T = 2,170$) and Culver-City ($T = 550$), a domain shift whose scenes follow a real road network rather than the default CARLA towns. The deployed selector is not retrained on either: the validate-trained model is applied unchanged, so this measures generalisation rather than re-fitting.

Table V reports both splits from the frozen replay. The policy ordering is identical to validate—fixed feature-level policies below Fixed L , the masked oracle marginally above it, each frozen selector between—and the transfer is in the channel use, not the F1: CA-TOSG spends 3.7–21.4% of the Fixed- F channel use on test and 2.5–18.4% on Culver-City, with realised F1 within 1.6 points of the masked oracle’s on test (98.4–99.1%) and 2.0 on Culver-City (98.0–99.4%).

The true end-to-end AP on the test and Culver-City splits under the frozen selector at $B_{\max} = 0.20$ is reported per SNR point in the committed product `true_e2e_ap_by_snr.csv`; its regime means are the Table II rows, on the pre-registered 11-point SNR grid with 200 paired realisations per point. Three things follow. First, the knee is a policy knee, not an AP knee, and sits at 10 dB, where the 16-QAM cliff clears: ρ_F jumps from 0 to 0.433 on test and 0 to 0.160 on Culver-City, flat thereafter. Second, that change buys a small AP gain: AP@0.5 moves 0.8691 $\rightarrow$ 0.8896 and AP@0.7 0.8027 $\rightarrow$ 0.8187 on test, and 0.7299 $\rightarrow$ 0.7474 and 0.6463 $\rightarrow$ 0.6556 on Culver-City—2.0 and 1.8 AP@0.5 points against 2.4 and 9.7 points of headroom, so what transfers is the communication saving rather than an AP gain. Third, the Rayleigh curves are flat at the Fixed- L point across 0–20 dB on both splits, the deep-fade frame BLER never falling low enough for feature transmission to be worthwhile. The non-monotonic dip previously reported at the 20 dB training-grid edge does not appear under the frozen selector—AP is flat above 10 dB on both splits—so the grid-edge artefact explanation is withdrawn.⁴

⁴The per-SNR curves of an earlier version of this section were produced by the retired global-sort scorer and its selector, not by the frozen selectors, and are withdrawn rather than restated. The frozen replacement (`end_to_end_ap_snr.py`) differs from the deployed AP engine in exactly one respect—the SNR/channel draw is pinned instead of sampled—and is gated on reproducing every committed row of the deployed table before any pinned number is produced. Coverage: test and Culver-City, AWGN and Rayleigh, all three frozen budgets; validate is not re-run here.

F. Is a Learned Selector Necessary? Comparison with an SNR-Threshold Rule

Since the channel features are the individually strongest cues (Table IV), we test whether a hand-tuned rule suffices. The supplementary comparison table sets RF feature subsets against a rule “if AWGN and $\hat{\gamma}_t > \tau$ then F else L ” (train on validate, evaluate on test). We report this honestly. Under cliff transport a retuned SNR threshold captures most of the attainable F1: on the payload-F1 frontier the learned RF and a tuned threshold sit almost on top of each other. As established in Section VI-A, the selector’s edge is not on F1: on test the nominal threshold rule attains the marginally higher realised F1 at every budget—0.89247, 0.89701 and 0.89900 against the selector’s 0.89148, 0.89691 and 0.89783—while spending $1.97\times$, $1.53\times$ and $1.47\times$ the channel use, and at $B_{\max} = 0.20$ and 0.30 doing so from outside the budget. Against the budget-matched τ_{feasible} the ordering reverses at one budget only: the selector is ahead by $+0.00067$ at $B_{\max} = 0.20$ and behind by -0.00174 and -0.00118 at 0.10 and 0.30 . What is pre-registered and survives is non-inferiority within a 0.005 margin at a 34.8% payload reduction — a figure that must be read with the comparator’s own budget in view: the re-tuned threshold spends 0.2168 Msym against the $B_{\max} = 0.20$ cap, i.e. it is over budget, so part of that reduction is the comparator’s overspend rather than the selector’s thrift. Against τ_{feasible} , the strictly budget-matched threshold of Section VI-F ($\tau = 13$, 0.1927 Msym), the reduction is 26.6% and the selector is ahead on F1 by $+0.00067$ rather than behind. Neither half of the input is sufficient on its own, and the shape of the failure is the informative part: given only the channel state the selector stops requesting features altogether at the two tighter budgets (feature-request rate 0, payload pinned at B_L), and at $B_{\max} = 0.30$, where it does activate the feature action, it is beaten on both axes at once—0.89529 against the full selector’s 0.89783 while spending $1.36\times$ the channel use (Section VI-C). We attribute the small accuracy margin to the sharp SNR cliff: under this transport the channel state carries most of the decision, and a one-line threshold on it therefore tracks the learned selector closely. That is a statement about this transport, not about sufficiency in general — the same cues become decisive as soon as the codec degrades gracefully (the supplementary material), and channel-side importance is not the same thing as channel-side sufficiency. The sentence this section settles on, and the one the rest of the paper is written to, is therefore: CA-TOSG does not win by raising average F1. A one-line SNR threshold and a three-scalar hand rule reach comparable F1, but they do so by sending more feature messages. What the task cues buy is knowing which of the frames the channel can carry are actually worth sending, so that perception stays non-inferior at lower communication. We therefore frame CA-TOSG as a lightweight, interpretable channel-aware selection policy whose aggregate operating point tracks a budget-matched

TABLE V

Generalisation of the validate-trained frozen selectors (no retraining) to the OPV2V test ($T = 2,170$ frames, scene-disjoint) and Culver-City ($T = 550$ frames, real-road-layout domain shift) splits. Every row is the mean over the same 200 paired CSI samplings of the frozen replay (SNR $\sim \mathcal{U}[0, 20]$ dB, 50/50 AWGN/Rayleigh); each frozen selector is paired with the SNR-threshold rule tuned to the same budget. Share is channel use as a fraction of Fixed F (0.990 Msym).

Policy	Mean F1	Channel use (Msym)	Share of Fixed F
OPV2V test ($T = 2,170$)			
Fixed L	0.8910	0.024	2.4%
Fixed F (LDPC + 16-QAM)	0.8483	0.990	100.0%
Fixed C_{256} (LDPC + 256-QAM)	0.8255	0.495	50.0%
CA-TOSG $B_{\max}=0.10 / \tau^*$	0.89148 / 0.89247	0.0368 / 0.0724	3.7% / 7.3%
CA-TOSG $B_{\max}=0.20 / \tau^*$	0.89691 / 0.89701	0.1414 / 0.2168	14.3% / 21.9%
CA-TOSG $B_{\max}=0.30 / \tau^*$	0.89783 / 0.89900	0.2120 / 0.3125	21.4% / 31.6%
Channel-aware oracle (masked)	0.9056	0.175	17.7%
OPV2V Culver-City ($T = 550$)			
Fixed L	0.8466	0.024	2.4%
Fixed F (LDPC + 16-QAM)	0.8099	0.990	100.0%
Fixed C_{256} (LDPC + 256-QAM)	0.7793	0.495	50.0%
CA-TOSG $B_{\max}=0.10 / \tau^*$	0.87230 / 0.87491	0.0283 / 0.0718	2.9% / 7.2%
CA-TOSG $B_{\max}=0.20 / \tau^*$	0.87355 / 0.88340	0.0423 / 0.2174	4.3% / 22.0%
CA-TOSG $B_{\max}=0.30 / \tau^*$	0.88286 / 0.88740	0.1601 / 0.3146	16.2% / 31.8%
Channel-aware oracle (masked)	0.8643	0.266	26.8%

threshold on this cliff codec closely, edging it on F1 while spending 26.6% less; the perception cues become decisive on graceful-degradation channels, where SNR is no longer a sufficient statistic and, in-distribution, the cue-based selector beats the best SNR threshold using the task cues alone (the supplementary material). The channel-averaged payloads are in Table III. Against the re-tuned SNR-threshold rule on test at $B_{\max} = 0.20$, the selector's F1 is lower than the nominal threshold rule by ≈ 0.0001 (95% CI upper bound -0.00002 , so still entirely below zero) yet within the pre-registered non-inferiority margin of 0.005, while reducing communication payload by 34.8% against a threshold that is itself over budget ($0.2168 > 0.20$ Msym), or 26.6% against the budget-matched τ_{feasible} . At $B_{\max} = 0.20$ and 0.30 the nominal threshold rule attains a marginally higher realised F1 (0.89701 and 0.89900 versus 0.89691 and 0.89783) at $1.53\times$ and $1.47\times$ the payload; against τ_{feasible} the ordering reverses at $B_{\max} = 0.20$ (0.89691 versus 0.89624).

G. Hand-Rule Baselines and Their Communication Cost

A hand rule with a second gate is the next challenger after a one-parameter threshold, and was pre-registered before it was run: $a_t = E$ if $d_t \leq \tau_E$, else F if $r_t = 1 - \text{BLER}_F(\hat{\gamma}_t, c_t) \geq \tau_F$, else L , with d_t one of three difficulty proxies from the same 21 cues in both orientations, both thresholds fitted on validate through the identical LOSO walk and budget constraint as the deployed selector.

At the two tighter budgets it cannot bid at all: the walk selects never- E and never- F at $B_{\max} = 0.10$ and 0.20 , degenerating to Fixed L , and at 0.30 the fitted reliability gate rediscovers the SNR threshold ($\rho_F = 0.29853$ against 0.29883 on validate). This is the four-rung ladder of Section VI-C: gating on the channel alone reaches only four mean payloads, and 0.10 and 0.20 fall between rungs.

Because a comparator that cannot enter the comparison is not a comparator, the arm was amended—in the record,

before running—so the feature gate is a conjunction: F when $d_t \geq \tau_D$ and $r_t \geq \tau_F$. That is three scalars, and we report it as three. It is feasible at every budget and at the primary cell it matches the selector on F1: 0.89697 against 0.89691, paired difference $+0.00005$ (95% CI $[-0.00002, +0.00012]$). No further comparator was run after seeing it.

The difference lies on the other axis: the hand rule spends 0.19900 Msym against 0.14141, a paired difference of $+0.05760$ (95% CI $[+0.05663, +0.05853]$)—40.7% more channel for that $+0.00005$ F1, or the selector reaching the same F1 with 28.9% less. The mechanism is selectivity per transmitted frame: over the cells each policy sends F on, the mean feature-over-object gain is 0.05022 (selector), 0.03660 (hand rule) and 0.03038 (SNR threshold)—the last indistinguishable from the 0.03042 unconditional mean, so the threshold's feature frames are no better chosen than average. The selector buys more F1 per transmitted frame and needs fewer of them; that is the payload gap, and it is what the ego-side cues contribute here.

Neither arm ever selects E : $\rho_E = 0.00000$ in every split $\times$ budget cell of both the two-scalar and the three-scalar rule, over all six cue orientations (Section VI-H).

H. Where the Gain Sits, and How the Policy Decides

The gain is frame-selective. Stratifying test frames by the ego's own object-level F1 under a reliable channel (AWGN 16 dB), the selector's gain over Fixed L rises monotonically with difficulty and reaches **+0.0660** F1 (95% CI $[+0.0591, +0.0730]$) on the hardest tercile; on the easy tercile of test it is negative (-0.00471 , 95% CI $[-0.00742, -0.00236]$ at $B_{\max} = 0.20$), where Fixed L already reaches 0.97962 and there is nothing left to buy—an over-request that a payload-penalised operating point would mitigate. The full stratification is in the supplementary material.

How the policy decides. Under AWGN the selector’s adaptation knee sits at the measured 10 dB where the 16-QAM frame-error cliff clears, and under Rayleigh it holds L across the whole range. It almost never takes the ego-only action— $\rho_E = 0.002$ on test against the oracle’s $\rho_E = 0.157$ under Rayleigh—which is a limitation of the trained policy rather than of the action set, and one no monotone hand rule repairs (Section VI-G). The two channel-side features carry 61.7% of the Gini importance against 38.3% spread over the 21 ego-side cues (Table IV); importance is not sufficiency, and the ablation shows the channel features alone cannot reach the full selector’s operating point (Section VI-C).

Communication–perception frontier. On the payload–F1 plane the three frozen selectors lie on the attainable frontier between Fixed L and the channel-aware oracle, while the fixed feature-level policies C_{16} and C_{256} sit far below it—high payload, low realised F1 under the LDPC cliff. The frontier figure and its λ -sweep construction are in the supplementary material.

I. Boundaries, Scale and Robustness

Cooperation can hurt. On 1.5 / 5.8 / 0.2% of validate / test / Culver-City frames the ego-only output strictly exceeds the object-level fused output, and on 1.3 / 6.5 / 0.9% the delivered feature message scores below the ego-only fallback. Two responses follow: an undeliverable feature action is removed from the feasible set by the 0.999 mask and the pipeline reverts to the ego-only output, while harm on a deliverable channel has no masking answer— E is already a deployed action with its own codepoint, so the remedy is a selection-policy question rather than a message-format one. The full account, including the easy-stratum cost, is in the supplementary material.

Collaborator scale. Requesting more collaborators has sharply diminishing returns: at $B_{\max} = 0.20$ on validate the second collaborator buys +0.0300 F1 for +0.1183 Msym and the third only +0.0061 for a further +0.0991 Msym. The budget does not transfer automatically: the selector is frozen against a per-frame budget measured with a single collaborator, and at $N = 3$ it spends 0.36842 Msym—already breached at $N = 2$ (0.26937)—so a deployment serving more collaborators must re-freeze or budget on the effective count k_{eff} .

Deployment robustness and cost. Three link imperfections degrade the selector only gracefully, because an uncertain channel state falls back to L : SNR-estimation noise up to $\sigma \approx 1$ dB costs ≤ 0.0002 F1, CSI aging under a Jakes model at 60 km/h costs -0.0025 , and acting on a one-frame-stale decision costs at most -0.0109 . The deployed Random Forest runs in 52.1 ± 5.6 ms ($P_{95} = 58.3$ ms) per frame on a single CPU core, the slowest of the three frozen selectors, within the 100 ms frame interval of a 10 Hz LiDAR cycle. Details and the full table are in the supplementary material, which also reports the Where2comm reference comparison (reference only: a different scorer, a perfect channel and a different

TABLE VI
Pre-registered sensitivity analyses (test at $B_{\max} = 0.20$ unless stated). Products: r23_scene_bootstrap.csv, r23_object_message_bler.csv, r23_fragmentation_harq.csv, r25_fragmentation_replay.csv.

Check	Quantity	Result
Scene-level bootstrap	$\Delta F1$ 95% CI	$[-0.001508, +0.001522]$ ($18.938 \times$)
	ΔB 95% CI	$[-0.119307, -0.029856]$ ($45.177 \times$)
$BLER_L = 0.10$	F1 cost, selector	-0.00634 (threshold -0.00588 , F
	payload	unchanged by construction
Fragmentation + HARQ	AWGN 8 dB BLER	$0.402400 \rightarrow 0.100363$ ($k=2$), 0.05
	payload factor	$1.226954 \times, 1.120770 \times$
	Rayleigh 0–20 dB	frame BLER 1.0 for every k

collaborator convention, so it is not ranked against the arms here).

Sensitivity analyses. Three pre-registered checks bound what the headline rests on, and none of them moves a conclusion (Table VI). Resampling the primary cell at the level of the 16 test scenes rather than the 200 CSI draws widens the F1 interval by $18.938 \times$ to $[-0.001508, +0.001522]$ and the payload interval by $45.177 \times$ to $[-0.119307, -0.029856]$ Msym, so the interval no longer excludes zero on F1 but does not reach the -0.005 margin either. Charging the object-level link with $BLER_L = 0.10$ costs the selector 0.00634 F1 against 0.00588 for the threshold rule and 0.00734 for Fixed L , at unchanged payload. Splitting the feature frame into $k \in \{2, 4\}$ fragments with one retransmission lowers the marginal AWGN cliff point at 8 dB from 0.402400 to 0.100363 and 0.057077, at 1.226954 $\times$ and 1.120770 $\times$ the payload, and does not open the Rayleigh branch at all.

VII. Conclusion

We presented CA-TOSG, a channel-aware task-oriented semantic granularity selector for V2V cooperative perception: from per-frame LiDAR cues, an estimated SNR and a channel-type indicator it outputs one mode of the deployed set $\mathcal{S} = \{E, L, F\}$ —ego-only E with no request, the compact object-level L , or the compressed feature-level F under 16-QAM (C_{256} is a physical-layer comparator, excluded by design, Section III-B). It is a lightweight Random Forest adding no learnable parameters to the perception pipeline; the contribution is the channel-aware granularity policy, of which the forest is one interpretable implementation (Section VI-F). It runs in 52.1 ± 5.6 ms ($P_{95} = 58.3$ ms) per frame on a single CPU core—the slowest of the three frozen selectors, since the deployment claim must hold for all of them—within the 100 ms budget of a 10 Hz LiDAR cycle (the supplementary material).

The results support three conclusions. First, under the evaluated LDPC + QAM transport, fixed feature-level communication is often dominated by compact object-level communication through channel-induced cliff effects. Second, the AP a granularity policy can contest is bounded by each split’s headroom (0.0550 validate, 0.0240 test, 0.0970 Culver-City) and the frozen selector converts at

most about a fifth of it (up to 21.3%, and 0.0% on Culver-City at the tightest budget), so its realised AP is reported descriptively; what transfers to the scene-disjoint test split and the Culver-City domain shift is the communication saving, 3.7–21.4% of fixed feature-level channel use across the three budgets on test. Third, the dominant decision signal is the channel state: a simple SNR-threshold rule tracks the learned selector closely on the channel-averaged payload–F1 frontier and in fact attains a marginally higher realised F1 at every budget, at $1.47\text{--}1.97\times$ the channel use; what the pre-registered comparison establishes is non-inferiority within a 0.005 margin at a 34.8% payload reduction against an over-budget nominal threshold — 26.6% against a budget-matched one — not dominance. The granularity policy’s gain over object-level communication is itself frame-selective, reaching $+0.0660$ F1 (95% CI $[+0.0591, +0.0730]$) on the hardest tercile of test frames under a reliable channel; we attribute the limited marginal value of the perception cues to the sharp LDPC + QAM SNR cliff, which makes channel state a near-sufficient statistic. The same boundary appears across detectors: repeated on a second backbone under an equal-budget controlled protocol, the procedure is in-sample effective but does not transfer, with the ego-only action collapsing further than on the mainline (the supplementary material). When the feature branch instead uses graceful importance-map JSCC, the estimated SNR becomes uninformative and the picture reverses: the best SNR threshold collapses to object-level performance, and, in-distribution, the cue-based selector beats it using the task cues alone (the supplementary material); a selector trained on one split and frozen does not yet transfer this graceful-channel gain across domains, which we leave to future work. Whether task-oriented perception cues are needed may thus be determined by the feature codec’s channel response: under the cliff codec evaluated here their contribution is to the shape of the policy rather than to peak accuracy, whereas the graceful-codec half of that statement rests on exploratory evidence gathered under the prior protocol and reported in the supplementary material, not re-evaluated under the frozen one.

References

- [1] R. Xu, H. Xiang, X. Xia, X. Han, J. Li, and J. Ma, “OPV2V: An open benchmark dataset and fusion pipeline for perception with vehicle-to-vehicle communication,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2022, pp. 2583–2589.
- [2] R. Xu, H. Xiang, Z. Tu, X. Xia, M.-H. Yang, and J. Ma, “V2X-ViT: Vehicle-to-everything cooperative perception with vision transformer,” in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2022, pp. 107–124.
- [3] C. Iliopoulos, A. Iossifides, C. H. Foh, and P. Chatzimisios, “IEEE 802.11bd for next-generation V2X communications: From protocol to services,” *IEEE Commun. Standards Mag.*, vol. 9, no. 2, pp. 88–98, Jun. 2025, doi: 10.1109/MCOM-STD.2025.3569015.
- [4] 3GPP, “Study on evaluation methodology of new V2X use cases for LTE and NR,” 3GPP, Tech. Rep. TR 37.885, 2019, rel-15, v15.3.0.
- [5] Y. Hu, S. Fang, Z. Lei, Y. Zhong, and S. Chen, “Where2comm: Communication-efficient collaborative perception via spatial confidence maps,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 35, 2022, pp. 4874–4886.
- [6] Y. Sheng, H. Ye, L. Liang, S. Jin, and G. Y. Li, “Semantic communication for cooperative perception based on importance map,” *Journal of the Franklin Institute*, vol. 361, no. 6, p. 106739, 2024.
- [7] J. Gan, Y. Sheng, H. Zhang, L. Liang, H. Ye, C. Guo, and S. Jin, “SComCP: Task-oriented semantic communication for collaborative perception,” *IEEE Trans. Veh. Technol.*, 2026, doi: 10.1109/TVT.2026.3654968.
- [8] L. Breiman, “Random forests,” *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.
- [9] Y. Han, H. Zhang, H. Li, Y. Jin, C. Lang, and Y. Li, “Collaborative perception in autonomous driving: Methods, datasets, and challenges,” *IEEE Intelligent Transportation Systems Magazine*, vol. 15, no. 6, pp. 131–151, 2023.
- [10] T.-H. Wang, S. Manivasagam, M. Liang, B. Yang, W. Zeng, and R. Urtasun, “V2vnet: Vehicle-to-vehicle communication for joint perception and prediction,” in *European Conference on Computer Vision (ECCV)*, 2020, pp. 605–621.
- [11] Y. Li, S. Ren, P. Wu, S. Chen, C. Feng, and W. Zhang, “Learning distilled collaboration graph for multi-agent perception,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021.
- [12] R. Xu, Z. Tu, H. Xiang, W. Shao, B. Zhou, and J. Ma, “Cobevt: Cooperative bird’s eye view semantic segmentation with sparse transformers,” in *Conference on Robot Learning (CoRL)*, 2022.
- [13] Y.-C. Liu, J. Tian, N. Glaser, and Z. Kira, “When2comm: Multi-agent perception via communication graph grouping,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 4106–4115.
- [14] Y. Lu, Y. Hu, Y. Zhong, D. Wang, Y. Wang, and S. Chen, “An extensible framework for open heterogeneous collaborative perception,” in *International Conference on Learning Representations (ICLR)*, 2024.
- [15] Y. Hu, X. Peng, S. Liu, Z. Gao, Y. Wang, and S. Chen, “Pragmatic communication in multi-agent collaborative perception,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [16] Q. Xie, X. Zhou, T. Qiu, Q. Zhang, and W. Qu, “Soft actor-critic-based multilevel cooperative perception for connected autonomous vehicles,” *IEEE Internet of Things Journal*, vol. 9, no. 21, pp. 21 370–21 381, 2022.
- [17] P. Liu, Z. Wang, G. Yu, B. Zhou, and P. Chen, “Region-based hybrid collaborative perception for connected autonomous vehicles,” *IEEE Trans. Veh. Technol.*, vol. 73, no. 3, pp. 3119–3128, 2024, doi: 10.1109/TVT.2023.3324439.
- [18] E. Bourtsoulatz, D. B. Kurka, and D. Gündüz, “Deep joint source-channel coding for wireless image transmission,” *IEEE Transactions on Cognitive Communications and Networking*, vol. 5, no. 3, pp. 567–579, 2019.
- [19] H. Xie, Z. Qin, G. Y. Li, and B.-H. Juang, “Deep learning enabled semantic communication systems,” *IEEE Transactions on Signal Processing*, vol. 69, pp. 2663–2675, 2021.
- [20] D. Gündüz, Z. Qin, I. E. Aguerri, H. S. Dhillon, Z. Yang, A. Yener, K. K. Wong, and C.-B. Chae, “Beyond transmitting bits: Context, semantics, and task-oriented communications,” *IEEE Journal on Selected Areas in Communications*, vol. 41, no. 1, pp. 5–41, 2023.
- [21] S. Yuan, J. Lu, X. Li, and H. Zhang, “Multi-mode task-oriented semantic communication for cooperative perception in IoT networks,” in *Proc. IEEE Int. Conf. Cloud Computing Technol. Sci. (CloudCom)*, 2025, pp. 1–8.
- [22] Y. Zhang, H. An, Z. Fang, G. Xu, Y. Zhou, X. Chen, and Y. Fang, “SmartCooper: Vehicular collaborative perception with adaptive fusion and judger mechanism,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024.
- [23] J. Shi, J. Zhao, L. Zhuo, X. Wang, X. Zhan, and H. Liu, “V2V cooperative perception with adaptive communication loss for autonomous driving,” *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 10, pp. 14 866–14 878, 2025, doi: 10.1109/TITS.2025.3582601.
- [24] J. Gan, L. Liang, H. Zhang, C. Guo, and S. Jin, “CoDS: Collaborative perception via digital semantic communication,” *arXiv preprint arXiv:2512.22513*, 2025, v1, 27 Dec 2025.
- [25] A. H. Lang, S. Vora, H. Caesar, L. Zhou, J. Yang, and O. Beijbom, “PointPillars: Fast encoders for object detection from point clouds,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 12 697–12 705.